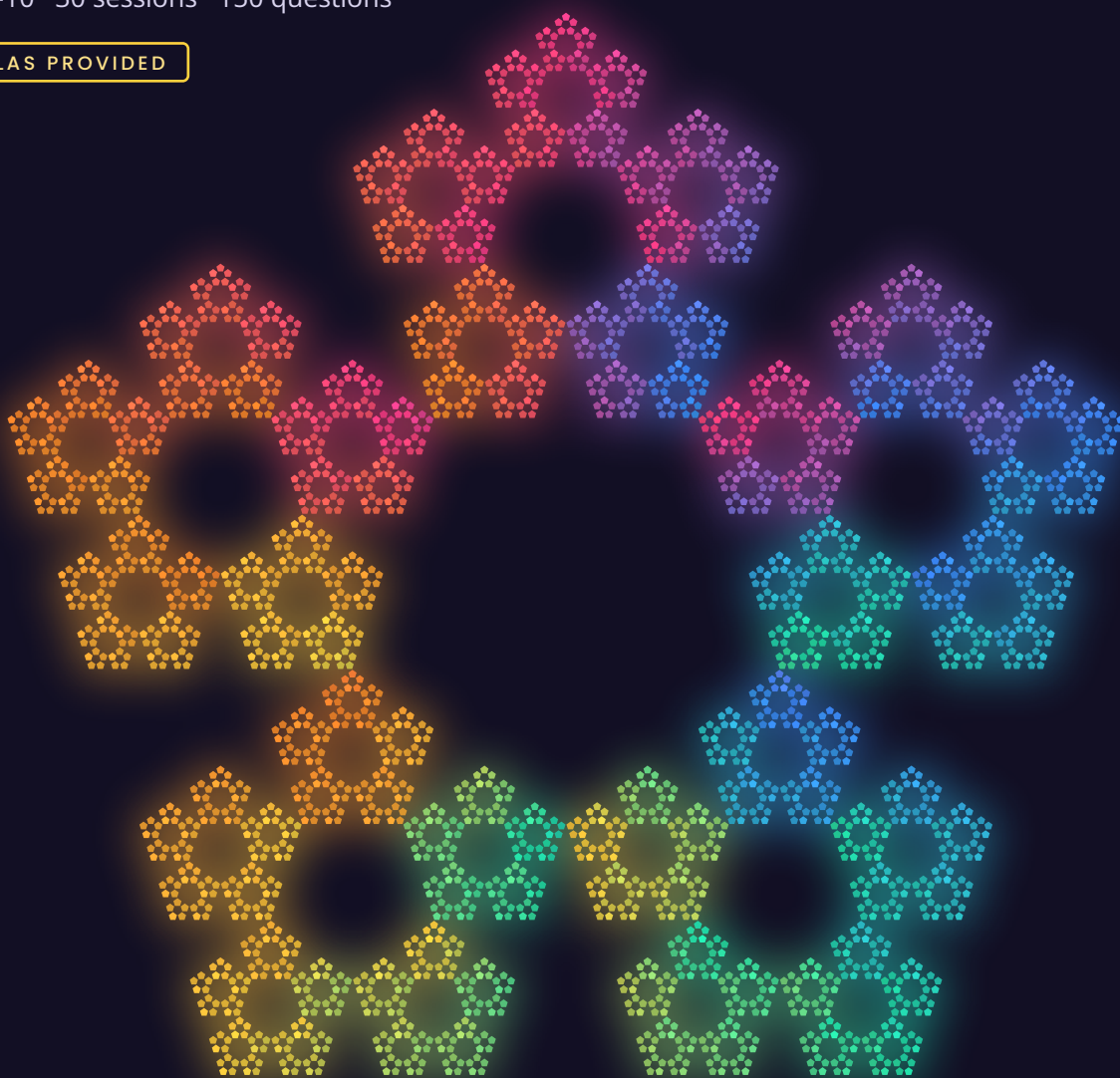


FIVE QUESTIONS. EVERY LESSON.

The Fractal Five

Weeks 1–10 · 30 sessions · 150 questions

FORMULAS PROVIDED



NAME _____

CLASS _____

SIERPIŃSKI PENTAGON

five copies of itself, each scaled by $1/\varphi^2 \approx 0.382$

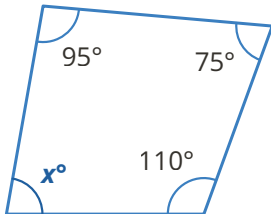
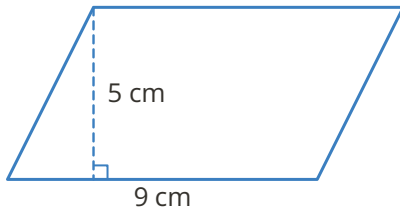
Week 1 · Session 1

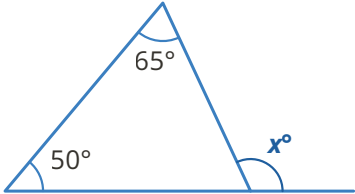
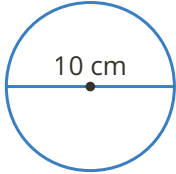
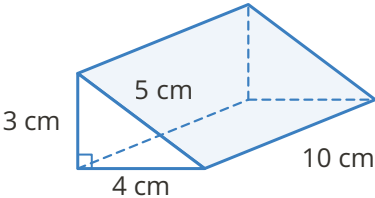
TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER						
4 Find the percentage increase in price. <table border="1" data-bbox="311 1541 601 1664"><tr><td>Original price</td><td>\$60</td></tr><tr><td>New price</td><td>\$75</td></tr></table>	Original price	\$60	New price	\$75	FORMULA $\% \text{ change} = \frac{\text{change}}{\text{original}} \times 100\%$ COPY SUBSTITUTE ANSWER		
Original price	\$60						
New price	\$75						
5 Find the range of these scores. <table border="1" data-bbox="146 1921 764 1986"><tr><td>Scores</td><td>12</td><td>7</td><td>15</td><td>9</td><td>20</td></tr></table>	Scores	12	7	15	9	20	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
Scores	12	7	15	9	20		

QUESTION	WORKING				
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY _____ SUBSTITUTE _____ _____ ANSWER _____				
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____				
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____				
4 Find the profit made on this sale. <table border="1" data-bbox="317 1543 596 1666"> <tr> <td>Cost price</td><td>\$120</td></tr> <tr> <td>Selling price</td><td>\$150</td></tr> </table>	Cost price	\$120	Selling price	\$150	FORMULA Profit = selling price – cost price COPY _____ SUBSTITUTE _____ _____ ANSWER _____
Cost price	\$120				
Selling price	\$150				
5 Find the probability that it does not rain. $P(\text{rain}) = 0.3$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____				

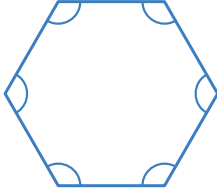
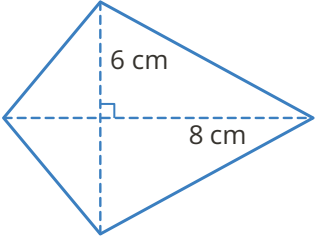
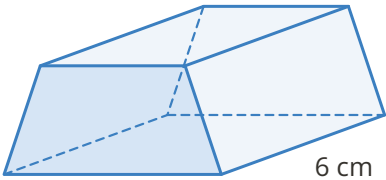
Week 1 · Session 3

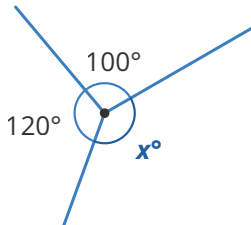
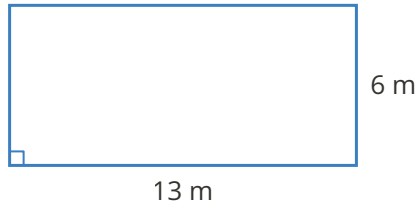
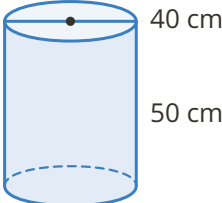
TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the sum of the interior angles of this hexagon. 	FORMULA Angle sum = $180^\circ \times (n - 2)$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this kite. 	FORMULA $A = \frac{1}{2}xy$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 The shaded face has an area of 15 cm^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Expand. $4(x + 3)$	FORMULA $a(b + c) = ab + ac$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = 4$. $y = 2x + 3$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER						
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY SUBSTITUTE ANSWER						
3 Find the capacity of this tank in litres, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$, $1000 \text{ cm}^3 = 1 \text{ L}$ COPY SUBSTITUTE ANSWER						
4 Write as a single power. $(2^3)^4$	FORMULA $(a^m)^n = a^{mn}$ COPY SUBSTITUTE ANSWER						
5 One marble is chosen at random from this bag. Find the probability that it is white. <table border="1" data-bbox="280 1912 628 2036"><tr><td>Colour</td><td>Red</td><td>White</td></tr><tr><td>Marbles</td><td>5</td><td>3</td></tr></table>	Colour	Red	White	Marbles	5	3	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER
Colour	Red	White					
Marbles	5	3					

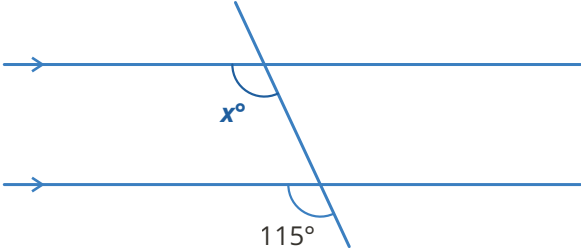
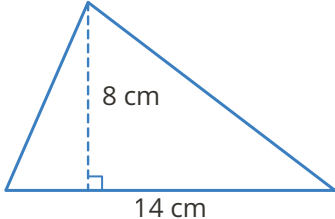
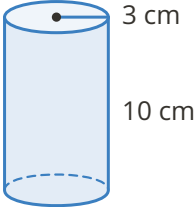
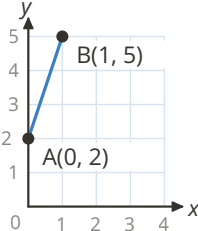
Week 2 · Session 2

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Corresponding angles on parallel lines are equal COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Calculate. 75% of \$48	FORMULA Amount = $\frac{\text{percentage}}{100} \times \text{quantity}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

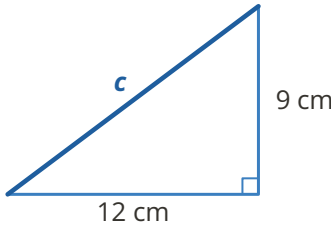
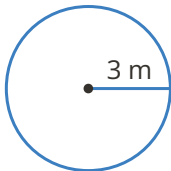
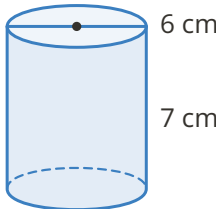
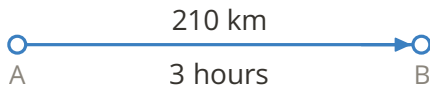
Week 2 · Session 3

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY _____ SUBSTITUTE _____ ANSWER _____						
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____						
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY _____ SUBSTITUTE _____ ANSWER _____						
4 A bus travels from A to B. Find the average speed. 	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY _____ SUBSTITUTE _____ ANSWER _____						
5 Find the mean of these goal tallies. <table border="1" data-bbox="150 1921 761 1986"><tr><td>Goals</td><td>7</td><td>9</td><td>4</td><td>10</td><td>5</td></tr></table>	Goals	7	9	4	10	5	FORMULA $\text{Mean} = \frac{\text{sum of scores}}{\text{number of scores}}$ COPY _____ SUBSTITUTE _____ ANSWER _____
Goals	7	9	4	10	5		

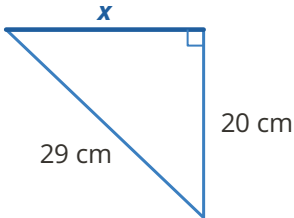
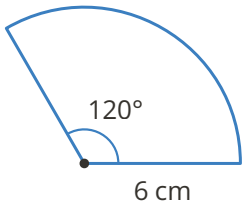
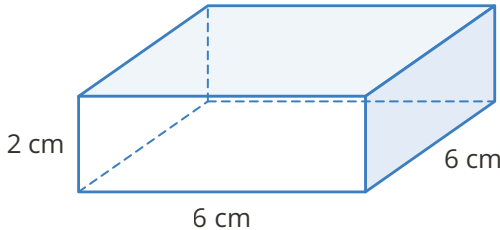
Week 3 · Session 1

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA $a^2 = c^2 - b^2$ COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this sector, correct to 1 decimal place. 	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the new population of the town. <table border="1" data-bbox="322 1541 590 1668"> <tr> <td>Population</td><td>2,400</td></tr> <tr> <td>Increase</td><td>5%</td></tr> </table>	Population	2,400	Increase	5%	FORMULA New amount = original $\times$ (100% + increase) COPY _____ SUBSTITUTE _____ ANSWER _____
Population	2,400				
Increase	5%				
5 Find the probability that the bus is not late. $P(\text{late}) = 0.15$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____				

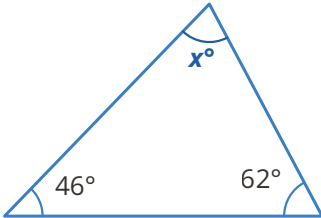
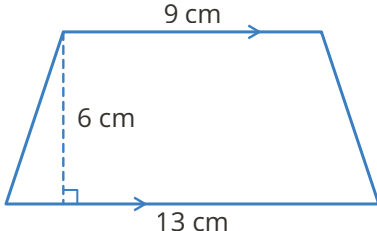
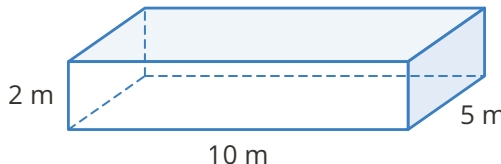
Week 3 · Session 2

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c = 180^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the area of this trapezium. 	FORMULA $A = \frac{1}{2}h(a + b)$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER						
4 Find the number. 40% of a number is 36	FORMULA Original amount = known amount $\div$ its percentage $\times 100$ COPY SUBSTITUTE ANSWER						
5 Find the range of these scores. <table><tr><td>Scores</td><td>35</td><td>48</td><td>29</td><td>52</td><td>41</td></tr></table>	Scores	35	48	29	52	41	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
Scores	35	48	29	52	41		

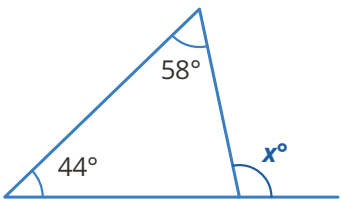
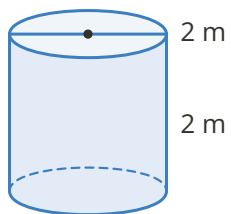
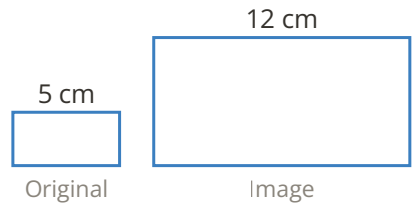
Week 3 · Session 3

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY _____ SUBSTITUTE _____ ANSWER _____
2 The area of this rectangle is 84 m^2. Find x. 	FORMULA $A = lw$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the capacity of this tank in litres, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$, $1 \text{ m}^3 = 1000 \text{ L}$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 The small rectangle is enlarged to make the large one. Find the scale factor. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = 2$. $y = 6x + 1$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____

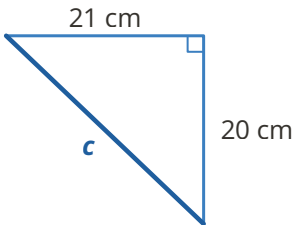
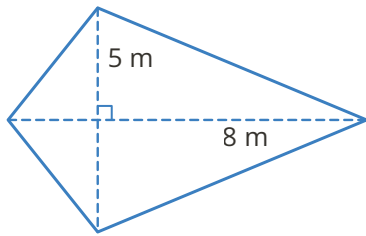
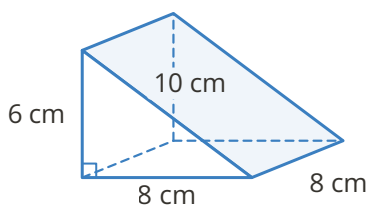
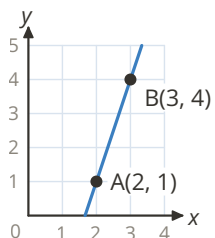
Week 4 · Session 1

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this kite. 	FORMULA $A = \frac{1}{2}xy$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 A train travels from A to B at an average speed of 60 km/h. Find x. 	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

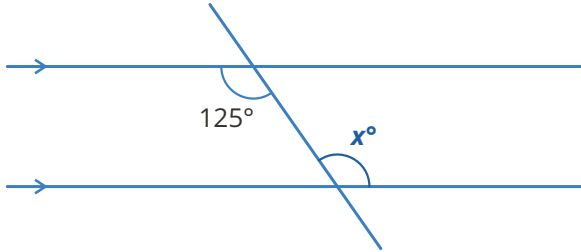
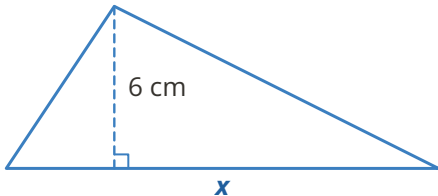
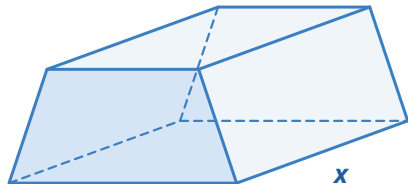
Week 4 · Session 2

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Alternate angles on parallel lines are equal COPY SUBSTITUTE ANSWER						
2 The area of this triangle is 48 cm^2. Find x. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER						
3 The shaded face has an area of 15 cm^2 and the volume is 90 cm^3. Find x. 	FORMULA $V = Ah$ COPY SUBSTITUTE ANSWER						
4 Calculate. 30% of 90 L	FORMULA Amount = $\frac{\text{percentage}}{100} \times \text{quantity}$ COPY SUBSTITUTE ANSWER						
5 One marble is chosen at random from this bag. Find the probability that it is black. <table border="1" data-bbox="280 1912 632 2036"><tr><td>Colour</td><td>Black</td><td>White</td></tr><tr><td>Marbles</td><td>7</td><td>3</td></tr></table>	Colour	Black	White	Marbles	7	3	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY SUBSTITUTE ANSWER
Colour	Black	White					
Marbles	7	3					

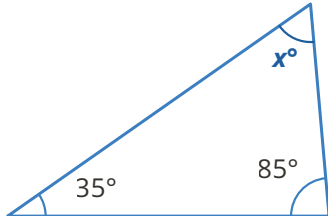
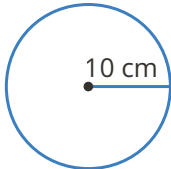
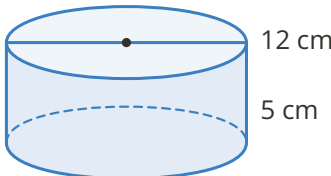
Week 4 · Session 3

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c = 180^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY SUBSTITUTE ANSWER						
4 Find the original price of the backpack. <table border="1" data-bbox="330 1541 580 1666"><tr><td>Discount</td><td>30%</td></tr><tr><td>Sale price</td><td>\$56</td></tr></table>	Discount	30%	Sale price	\$56	FORMULA Original amount = known amount $\div$ its percentage $\times 100$ COPY SUBSTITUTE ANSWER		
Discount	30%						
Sale price	\$56						
5 Find the mean of these scores. <table border="1" data-bbox="146 1921 764 1986"><tr><td>Scores</td><td>13</td><td>19</td><td>16</td><td>10</td><td>17</td></tr></table>	Scores	13	19	16	10	17	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Scores	13	19	16	10	17		

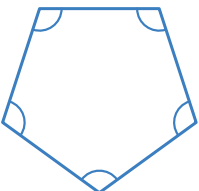
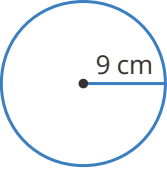
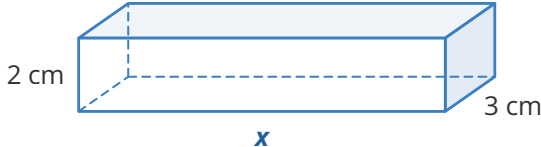
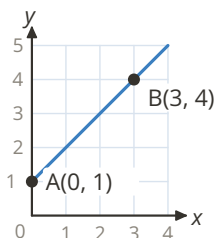
Week 5 · Session 1

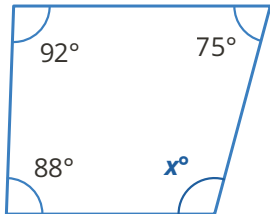
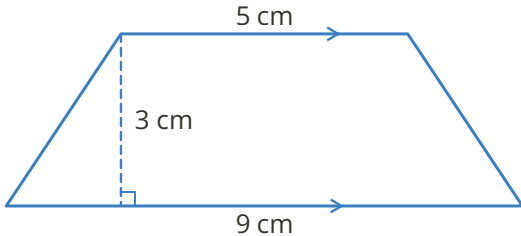
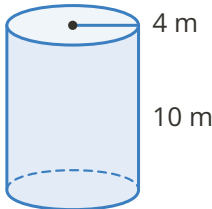
TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the sum of the interior angles of this pentagon. 	FORMULA Angle sum = $180^\circ \times (n - 2)$ COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 The volume of this rectangular prism is 60 cm^3. Find x. 	FORMULA $V = lwh$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the percentage increase in price. <table border="1" data-bbox="311 1545 598 1668"> <tr> <td>Original price</td><td>\$50</td></tr> <tr> <td>New price</td><td>\$60</td></tr> </table>	Original price	\$50	New price	\$60	FORMULA % change = $\frac{\text{change}}{\text{original}} \times 100\%$ COPY _____ SUBSTITUTE _____ ANSWER _____
Original price	\$50				
New price	\$60				
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____				

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the area of this trapezium. 	FORMULA $A = \frac{1}{2}h(a + b)$ COPY SUBSTITUTE ANSWER						
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER						
4 Find the profit made on this sale. <table><tr><td>Cost price</td><td>\$200</td></tr><tr><td>Selling price</td><td>\$260</td></tr></table>	Cost price	\$200	Selling price	\$260	FORMULA Profit = selling price – cost price COPY SUBSTITUTE ANSWER		
Cost price	\$200						
Selling price	\$260						
5 Find the range of these marks. <table><tr><td>Marks</td><td>61</td><td>47</td><td>55</td><td>70</td><td>52</td></tr></table>	Marks	61	47	55	70	52	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
Marks	61	47	55	70	52		

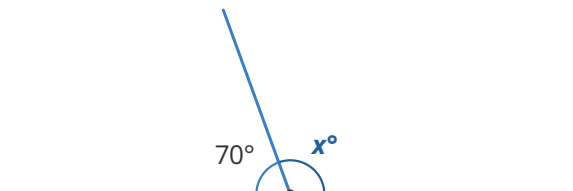
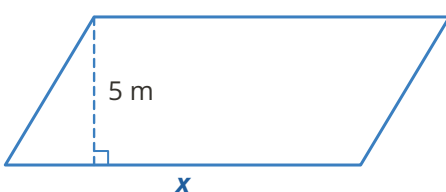
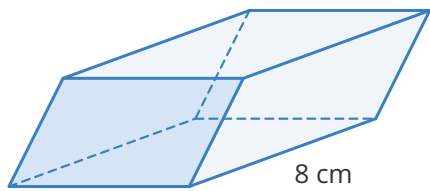
Week 5 · Session 3

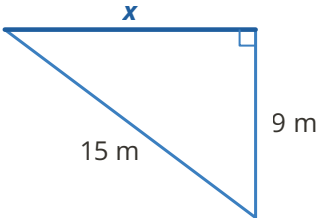
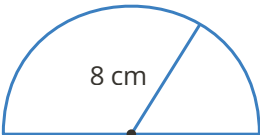
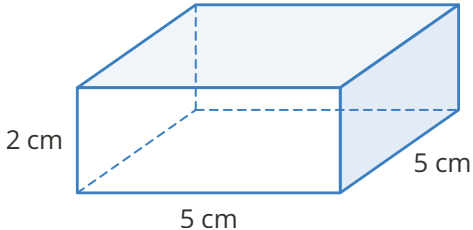
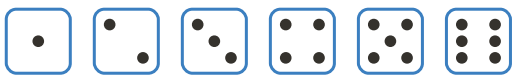
TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA Angles on a straight line add to 180° COPY _____ SUBSTITUTE _____ ANSWER _____				
2 The area of this parallelogram is 60 m^2. Find x. 	FORMULA $A = bh$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 The shaded face has an area of 15 cm^2. Find the volume of the prism. 	FORMULA $V = Ah$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the sale price. <table border="1" data-bbox="311 1534 598 1668"> <tr> <td>Original price</td><td>\$45</td></tr> <tr> <td>Discount</td><td>20%</td></tr> </table>	Original price	\$45	Discount	20%	FORMULA New amount = original $\times$ (100% – decrease) COPY _____ SUBSTITUTE _____ ANSWER _____
Original price	\$45				
Discount	20%				
5 Find the probability that the counter is not red. $P(\text{red}) = \frac{1}{4}$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____				

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a^2 = c^2 - b^2$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
2 Find the area of this semicircle, correct to 1 decimal place. 	FORMULA $A = \frac{1}{2}\pi r^2$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
4 Evaluate. 7^0	FORMULA $a^0 = 1$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
5 A fair die is rolled. Find the probability of rolling a 5. 	FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____

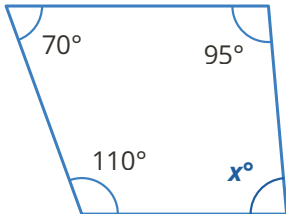
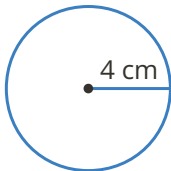
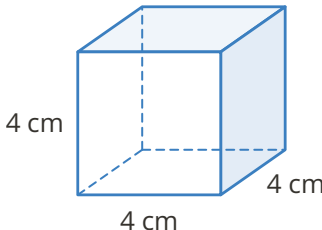
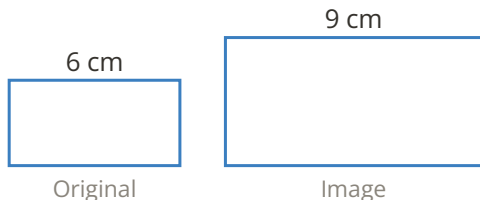
Week 6 · Session 2

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = 2\pi r$ COPY SUBSTITUTE ANSWER						
3 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER						
4 The small rectangle is enlarged to make the large one. Find the scale factor. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER						
5 Find the mean of these scores. <table><tr><td>Scores</td><td>8</td><td>3</td><td>5</td><td>6</td><td>8</td></tr></table>	Scores	8	3	5	6	8	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Scores	8	3	5	6	8		

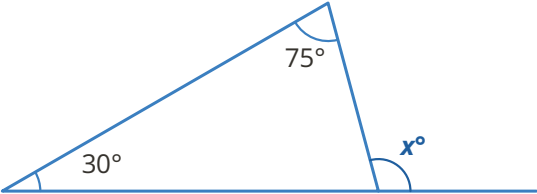
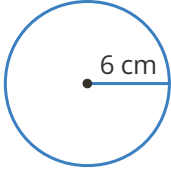
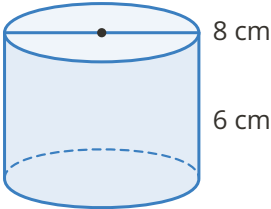
Week 6 · Session 3

TERM 2

Copy the formula, substitute, then solve.

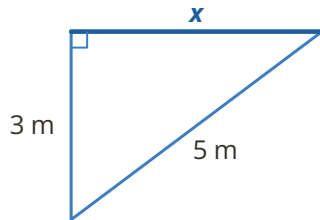
NAME _____

DATE _____

QUESTION	WORKING
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this circle, correct to 1 decimal place. 	FORMULA $A = \pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Expand. $4(m + 9)$	FORMULA $a(b + c) = ab + ac$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = 1$. $y = 10x - 4$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____

QUESTION

- 1 Find the value of x .



WORKING

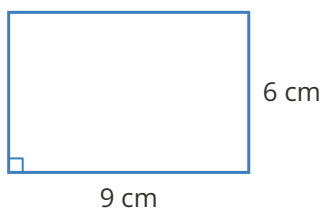
FORMULA $a^2 = c^2 - b^2$

COPY

SUBSTITUTE

ANSWER

- 2 Find the area of this rectangle.



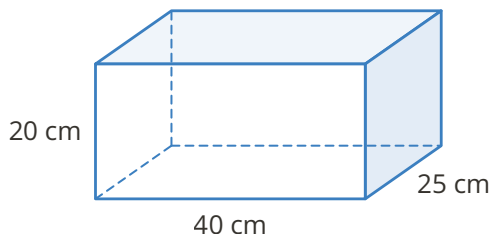
FORMULA $A = lw$

COPY

SUBSTITUTE

ANSWER

- 3 Find the capacity of this tank in litres.



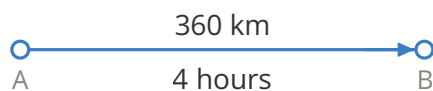
FORMULA $V = lwh$, $1000 \text{ cm}^3 = 1 \text{ L}$

COPY

SUBSTITUTE

ANSWER

- 4 A train travels from A to B. Find the average speed.



FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$

COPY

SUBSTITUTE

ANSWER

- 5 One marble is chosen at random from this bag. Find the probability that it is blue.

Colour	Blue	Green	Red
Marbles	6	4	2

FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$

COPY

SUBSTITUTE

ANSWER

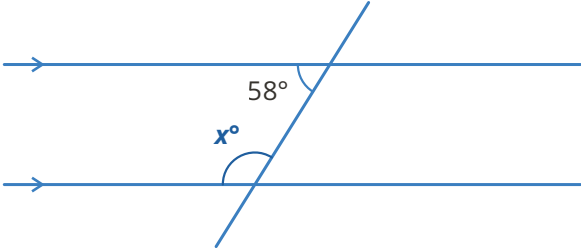
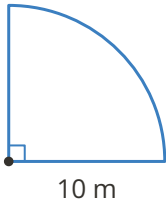
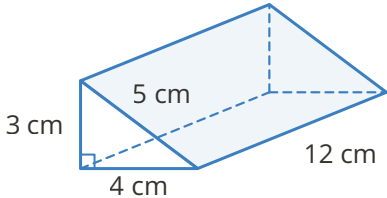
Week 7 · Session 2

TERM 2

Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING				
1 Find the value of x. 	FORMULA Co-interior angles on parallel lines add to 180° COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this quadrant, correct to 1 decimal place. 	FORMULA $A = \frac{\theta}{360} \times \pi r^2$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the profit made on this sale. <table border="1" data-bbox="317 1543 596 1666"> <tr> <td>Cost price</td><td>\$240</td></tr> <tr> <td>Selling price</td><td>\$300</td></tr> </table>	Cost price	\$240	Selling price	\$300	FORMULA Profit = selling price – cost price COPY _____ SUBSTITUTE _____ ANSWER _____
Cost price	\$240				
Selling price	\$300				
5 Find the probability that it does not snow. $P(\text{snow}) = 0.08$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____				

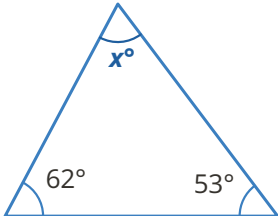
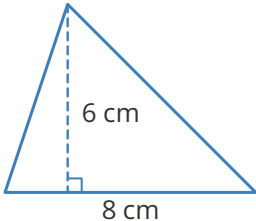
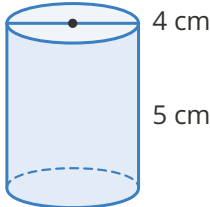
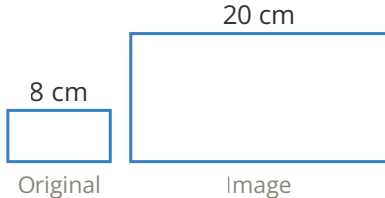
Week 7 · Session 3

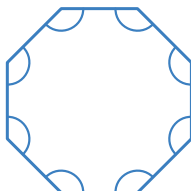
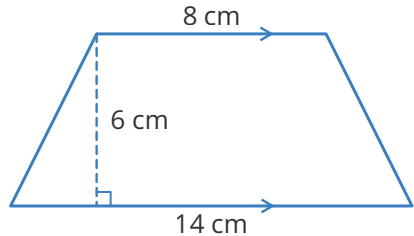
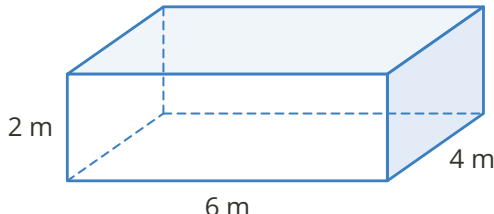
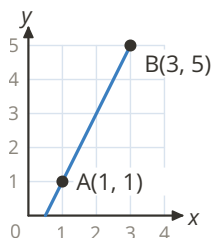
TERM 2

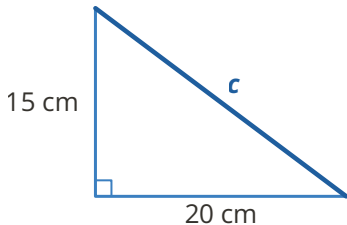
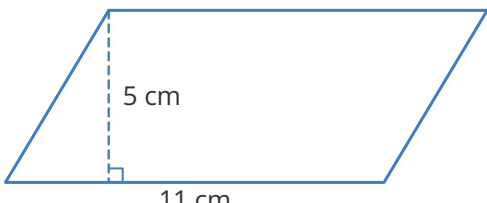
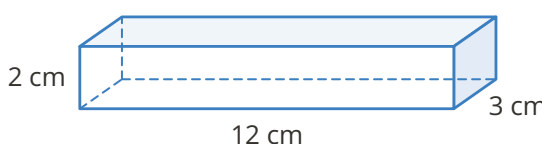
Copy the formula, substitute, then solve.

NAME _____

DATE _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA $a + b + c = 180^\circ$ COPY SUBSTITUTE ANSWER						
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY SUBSTITUTE ANSWER						
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY SUBSTITUTE ANSWER						
4 The small rectangle is enlarged to make the large one. Find the scale factor. 	FORMULA Scale factor = $\frac{\text{image length}}{\text{original length}}$ COPY SUBSTITUTE ANSWER						
5 Find the range of these scores. <table><tr><td>Scores</td><td>9</td><td>3</td><td>12</td><td>7</td><td>5</td></tr></table>	Scores	9	3	12	7	5	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
Scores	9	3	12	7	5		

QUESTION	WORKING
1 Find the sum of the interior angles of this octagon. 	FORMULA Angle sum = $180^\circ \times (n - 2)$ COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this trapezium. 	FORMULA $A = \frac{1}{2}h(a + b)$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Calculate. 12% of \$50	FORMULA Amount = $\frac{\text{percentage}}{100} \times \text{quantity}$ COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ ANSWER _____

QUESTION	WORKING						
1 Find the length of the hypotenuse, c. 	FORMULA $c^2 = a^2 + b^2$ COPY SUBSTITUTE ANSWER						
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER						
3 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY SUBSTITUTE ANSWER						
4 Expand. $5(x + 2)$	FORMULA $a(b + c) = ab + ac$ COPY SUBSTITUTE ANSWER						
5 Find the mean of these marks. <table><tr><td>Marks</td><td>45</td><td>52</td><td>48</td><td>50</td><td>55</td></tr></table>	Marks	45	52	48	50	55	FORMULA Mean = $\frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Marks	45	52	48	50	55		

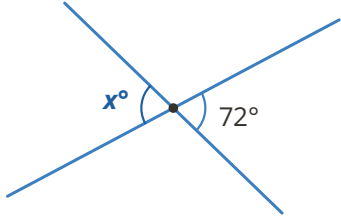
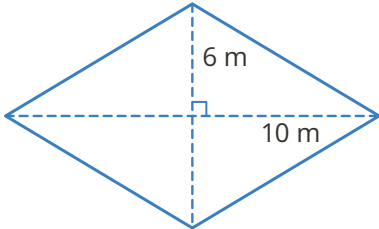
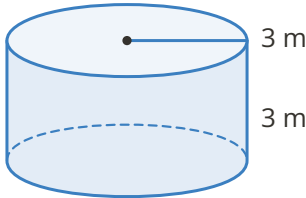
Week 8 · Session 3

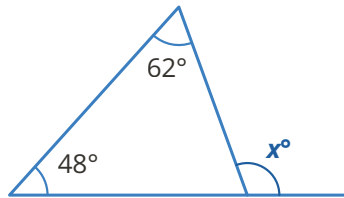
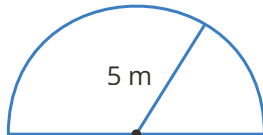
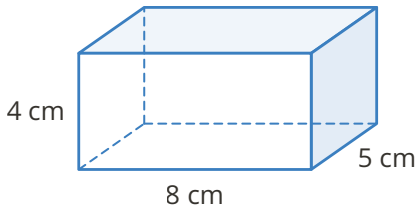
TERM 2

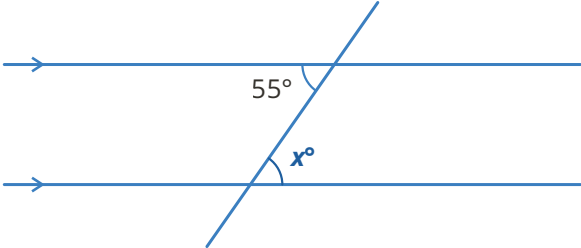
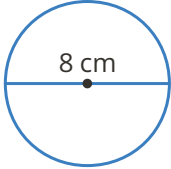
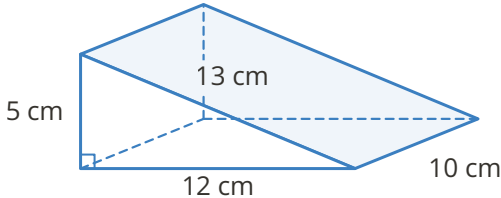
Copy the formula, substitute, then solve.

NAME _____

DATE _____

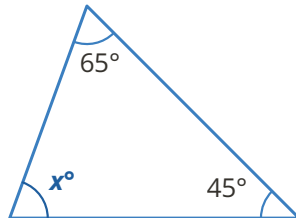
QUESTION	WORKING
1 Find the value of x. 	FORMULA Vertically opposite angles are equal COPY _____ SUBSTITUTE _____ ANSWER _____
2 Find the area of this rhombus. 	FORMULA $A = \frac{1}{2}xy$ COPY _____ SUBSTITUTE _____ ANSWER _____
3 Find the surface area of this closed cylinder, correct to 1 decimal place. 	FORMULA $SA = 2\pi r^2 + 2\pi rh$ COPY _____ SUBSTITUTE _____ ANSWER _____
4 Find the new mass. Decrease 250 kg by 20%	FORMULA New amount = original $\times$ (100% – decrease) COPY _____ SUBSTITUTE _____ ANSWER _____
5 Find the value of y when $x = 4$. $y = 3x + 1$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Exterior angle = sum of the two interior opposite angles COPY SUBSTITUTE ANSWER						
2 Find the area of this semicircle, correct to 1 decimal place. 	FORMULA $A = \frac{1}{2}\pi r^2$ COPY SUBSTITUTE ANSWER						
3 Find the capacity of this container in millilitres. 	FORMULA $V = lwh$, $1 \text{ cm}^3 = 1 \text{ mL}$ COPY SUBSTITUTE ANSWER						
4 Write as a single power. $3^2 \times 3^5$	FORMULA $a^m \times a^n = a^{m+n}$ COPY SUBSTITUTE ANSWER						
5 Find the range of these heights. <table><tr><td>cm</td><td>104</td><td>96</td><td>118</td><td>101</td><td>99</td></tr></table>	cm	104	96	118	101	99	FORMULA Range = highest – lowest COPY SUBSTITUTE ANSWER
cm	104	96	118	101	99		

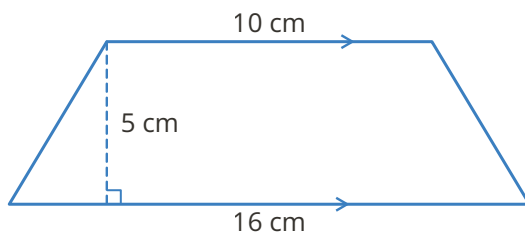
QUESTION	WORKING				
1 Find the value of x. 	FORMULA Alternate angles on parallel lines are equal COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the circumference of this circle, correct to 1 decimal place. 	FORMULA $C = \pi d$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the surface area of this triangular prism. 	FORMULA $SA = 2 \times \text{area of triangle} + \text{area of the 3 rectangles}$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the price before the increase. <table border="1" data-bbox="331 1543 582 1668"> <tr> <td>Increase</td><td>5%</td></tr> <tr> <td>New price</td><td>\$252</td></tr> </table>	Increase	5%	New price	\$252	FORMULA Original amount = known amount $\div$ its percentage $\times 100$ COPY _____ SUBSTITUTE _____ ANSWER _____
Increase	5%				
New price	\$252				
5 Find the probability that a six is not rolled. $P(\text{six}) = \frac{1}{6}$	FORMULA $P(\text{not } E) = 1 - P(E)$ COPY _____ SUBSTITUTE _____ ANSWER _____				

QUESTION

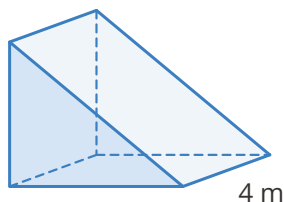
- 1** Find the value of x .



- 2** Find the area of this trapezium.



- 3** The shaded face has an area of 15 m^2 . Find the volume of the prism.



- 4** Find the percentage increase in price.

Original price	\$25
New price	\$30

- 5** One marble is chosen at random from this bag. Find the probability that it is red.

Colour	Red	Yellow	Blue
Marbles	2	3	5

WORKING

FORMULA $a + b + c = 180^\circ$

COPY

SUBSTITUTE

ANSWER

FORMULA $A = \frac{1}{2}h(a + b)$

COPY

SUBSTITUTE

ANSWER

FORMULA $V = Ah$

COPY

SUBSTITUTE

ANSWER

FORMULA $\% \text{ change} = \frac{\text{change}}{\text{original}} \times 100\%$

COPY

SUBSTITUTE

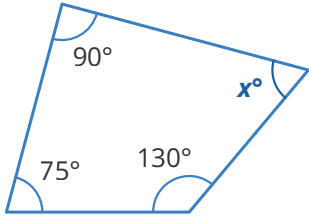
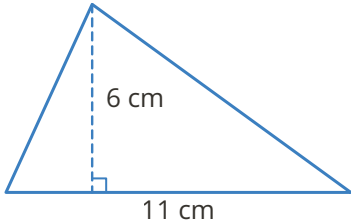
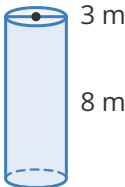
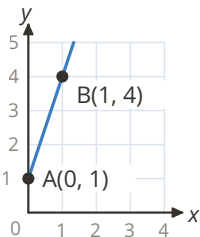
ANSWER

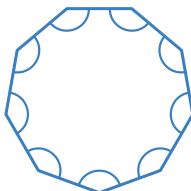
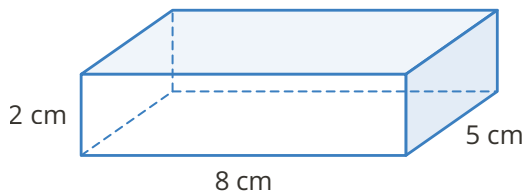
FORMULA $P(E) = \frac{\text{favourable outcomes}}{\text{total outcomes}}$

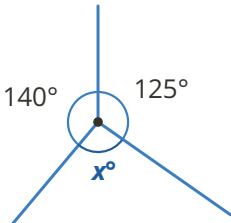
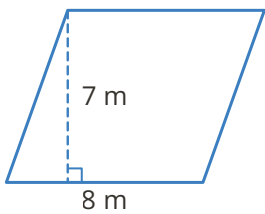
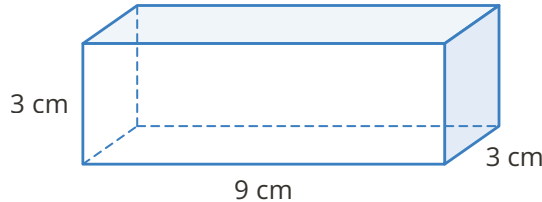
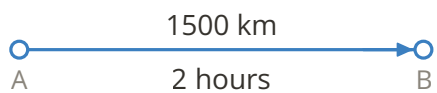
COPY

SUBSTITUTE

ANSWER

QUESTION	WORKING
1 Find the value of x. 	FORMULA $a + b + c + d = 360^\circ$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
2 Find the area of this triangle. 	FORMULA $A = \frac{1}{2}bh$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
3 Find the volume of this cylinder, correct to 1 decimal place. 	FORMULA $V = \pi r^2 h$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
4 Write as a single power. $6^5 \div 6^3$	FORMULA $a^m \div a^n = a^{m-n}$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____
5 Find the gradient of line AB. 	FORMULA $m = \frac{\text{rise}}{\text{run}}$ COPY _____ SUBSTITUTE _____ _____ ANSWER _____

QUESTION	WORKING				
1 Find the sum of the interior angles of this nonagon. 	FORMULA Angle sum = $180^\circ \times (n - 2)$ COPY _____ SUBSTITUTE _____ ANSWER _____				
2 Find the area of this rectangle. 	FORMULA $A = lw$ COPY _____ SUBSTITUTE _____ ANSWER _____				
3 Find the volume of this rectangular prism. 	FORMULA $V = lwh$ COPY _____ SUBSTITUTE _____ ANSWER _____				
4 Find the new price. <table border="1" data-bbox="311 1534 598 1668"> <tr> <td>Original price</td><td>\$150</td></tr> <tr> <td>Increase</td><td>30%</td></tr> </table>	Original price	\$150	Increase	30%	FORMULA New amount = original $\times$ (100% + increase) COPY _____ SUBSTITUTE _____ ANSWER _____
Original price	\$150				
Increase	30%				
5 Find the value of y when $x = 5$. $y = 2x - 3$	FORMULA $y = mx + c$ COPY _____ SUBSTITUTE _____ ANSWER _____				

QUESTION	WORKING						
1 Find the value of x. 	FORMULA Angles at a point add to 360° COPY SUBSTITUTE ANSWER						
2 Find the area of this parallelogram. 	FORMULA $A = bh$ COPY SUBSTITUTE ANSWER						
3 Find the surface area of this prism. 	FORMULA $SA = 2(lw + lh + wh)$ COPY SUBSTITUTE ANSWER						
4 A plane travels from A to B. Find the average speed. 	FORMULA $\text{Speed} = \frac{\text{distance}}{\text{time}}$ COPY SUBSTITUTE ANSWER						
5 Find the mean of these goal tallies. <table border="1" data-bbox="150 1912 754 1980"><tr><td>Goals</td><td>6</td><td>2</td><td>9</td><td>4</td><td>9</td></tr></table>	Goals	6	2	9	4	9	FORMULA $\text{Mean} = \frac{\text{sum of scores}}{\text{number of scores}}$ COPY SUBSTITUTE ANSWER
Goals	6	2	9	4	9		